AI-Optical Design Evolution
- Four Pillars -

Summary Table with Key References

with the Classical-Asymptote Test for Quantum Photonics

Updated April 9, 2026 — v14

Pillar 1: 湧現 (Emergence)
Pillar 2: 湧創 (Continuous Innovation)
Pillar 3: 湧視 (Unaddressed Limits)
Pillar 4: 湧知 (Emergent Knowledge)

Original three pillars: **湧現** | **湧創** | **湧視**

Proposed fourth pillar: **湧知**

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1 Pillar 1: 湧現 (Emergence)

Pillar 1: 湧現 — Emergence

"Collective behavior of optical elements creates new functionalities not inherent in any single component."

#	Emergent System	Components	Emergent Behavior	Status (May 2025)	Status (Mar 2026)	Ref.
1	Computational Imaging	Coded aperture + Algorithm + Detector	Super-resolution beyond diffraction limit	Lab demos	**AI wavefront compensation: 40% → 98% accuracy recovery**	R19; R20
2	Metalens Arrays	Subwavelength structures	Anomalous dispersion control	Active research	**Achromatic RGB bilayer metalens: 80° FOV, NA=0.65**	R17
3	Adaptive Imaging	Deformable mirror + WFS + Control	Real-time aberration correction	Established	**DRL auto-tuning of photonic circuits in 21 iterations**	R9
4	Light Field Cameras	Microlens array + Sensor + Processing	Post-capture refocusing	Commercial	Mature; incremental	—
5	Quantum Imaging	Entangled photons + Coincidence detection	Ghost imaging, sub-shot noise	Early lab demos	**Metasurface SPDC: $>10^4\times$ resolution cells vs bulk crystal**	R2
6	**Quantum-Meta-surface Multi-modal (NEW)**	Pol.-entangled source + Metasurface + ICCD	Phase contrast + edge + superimposed; self-calibration	Not identified	**Self-calibrated edge imaging via "off-state" noise subtraction**	R3
7	**AI-Optical Co-Design (NEW)**	LLM + Physics engine + RL + Zemax	Lens systems from natural language specs	Concept only	**OptiAgent: first LLM-agent for optical design with physics-driven RL**	R1
8	**Nonlinear Meta-surface Quantum Sources (NEW)**	Metagrating on $LiNbO_3$ thin film	All-optical scanning of entangled photon emission angle	Not identified	**Demonstrated at IR wavelengths**	R2

2 Pillar 2: 湧創 (Continuous Innovation)

Pillar 2: 湧創 — Continuous Innovation

"Shifting from periodic breakthroughs to constant incremental improvements through real-time feedback loops."

#	Aspect	Discrete Evolution (2025)	Continuous Target (2025)	Current Status (Mar 2026)	Ref.
1	Design Cycle	6–24 months per design	Real-time adaptation (hours/days)	**Metasurface: hours (achieved); Classical: days (partial); Hybrid: weeks**	R5; R1
2	Feedback Integration	Post-production analysis	In-situ performance monitoring	**ML tolerance prediction replaces millions of ray-trace perturbations; "right first time"**	R11
3	Knowledge Transfer	Conference papers, patents	Shared AI models, cloud databases	**Open datasets: OptiDesignQA; Open tools: LensAI, LensRL; No unified platform yet**	R14; R15
4	Optimization	Fixed merit functions	Dynamic merit function evolution	**Still static; RL reward shaping (OptiAgent) closest but not self-evolving**	R1
5	Manufacturing Link	Sequential handoff	Concurrent design-manufacture	**Micro-lens + LED co-design in production (CES 2026); digital twin partial**	R12
6	**AI Agent (NEW)**	Not in original	—	**LLM agents with Zemax function calls in closed loop; RL on H100 GPUs**	R1; R18
7	**Fab-Aware Design (NEW)**	Not in original	—	**ML-designed lens: 90% prob. of meeting spec despite tolerances**	R11

3 Pillar 3: 湧視 (Unaddressed Limits)

Pillar 3: 湧視 — Unaddressed Physical Limits

"Physical phenomena that current lens design paradigms ignore, work around, or haven't yet exploited."

#	Phenom.	2025 Assess.	Untapped Potential	Status (Mar 2026)	Maturity Shift	Ref.
1	Quantum Noise	Seen as limitation	Information source	**Entangled photon pairs from metasurface SPDC; improved SNR**	Limitation → Resource	R2; R4
2	Thermodynamic Limits	Ignored	Energy optimization	**Still largely ignored; thermal meta-devices advancing separately**	No change	R22
3	Vacuum Fluctuations	Not considered	Casimir effects	**Remains exotic; no integration**	No change	—
4	Nonlinear Dispersion	Avoided	Soliton benefits	**AI navigates complex parameter spaces; OptoGPT for multilayer**	Minimized → AI-navigated	R10
5	Spin-Orbit Coupling	Unknown	Geometric phase control	**Photonic skyrmions via GRIN + metasurface; vdW (8 μm hBN) vortex gen.**	Unknown → Active	R6; R7
6	Topology	Not exploited	Protected optical states	**Fab-compatible SOI spin-valley Hall phase; pseudo-spin framework**	Not Exploited → Design Tool	R8
7	Time Reversal	Static only	Phase conjugation	**In-situ RL for holographic phase; adaptive control advancing**	Static → Adaptive	R9

4 The Classical-Asymptote Test: A Formal Criterion for 湧視

Classical-Asymptote Test — Quantifying 湧視

Motivated by Barrett's argument on the inevitability of quantum computing (R23), we formalize an analogous inevitability criterion for quantum-photonic capabilities. The test sharpens 湧視 from a qualitative category into a falsifiable property.

A quantum-photonic topic is said to *clear the bar* if it satisfies all four conditions:

(i) **Classical asymptote.** A rigorous figure of merit exists on which the best classical implementation is provably zero or bounded by physics that no amount of engineering can cross (e.g. shot-noise floor, no-cloning, computational complexity).

(ii) **Scaling pathway.** A demonstrated or credible hardware roadmap scales the quantum resource monotonically (qubits, squeezing dB, entangled flux, memory lifetime).

(iii) **Algorithmic / design velocity.** Algorithms, protocols, or AI-assisted designs improve faster than hardware — a dynamic feedback loop that closes the capability gap from above.

(iv) **Discovery → design shift.** The capability transitions its field from iterative trial-and-error to deterministic engineering.

This criterion is strictly more informative than Barrett's implicit test because it separates *physics inevitability* (conditions i–ii) from *engineering inevitability* (conditions iii–iv). A topic may be physics-inevitable without being engineering-inevitable, or vice versa. Only topics that satisfy all four are candidates for "inevitable" in the Barrett sense.

Application to 18 quantum-photonic topics. Table 1 applies the test to a representative set of topics spanning quantum communication, metrology, imaging, sensing, and computation. Each topic is mapped to its most relevant pillar(s) and scored as Pass (clears all four conditions), Cusp (clears i–ii but not iii–iv), or Weak (contested or failing).

Table 1: Classical-Asymptote Test applied to 18 quantum-photonic topics. The Bar is decomposed into **Physics** (conditions i–ii: classical asymptote + scaling pathway) and **Engineering** (conditions iii–iv: algorithmic velocity + discovery→design shift). A topic is Barrett-inevitable only if both columns are Pass.

#	Topic	Key Reference	Pillar	Phys. (i–ii)	Eng. (iii–iv)
1	BB84 QKD & deployed metro networks	R24; ID Quantique commercial	湧視/湧現	**Pass**	**Pass**
2	Satellite QKD (Micius)	R25; R26	湧視/湧現	**Pass**	**Pass**
3	Twin-field QKD (beats PLOB bound)	Lucamarini et al., *Nature* 557, 400 (2018)	湧視	**Pass**	Cusp
4	Device-independent QKD	Nadlinger et al., *Nature* 607, 682 (2022); Zhang et al., *Nature* 607, 687 (2022)	湧視	**Pass**	Cusp
5	Squeezed-light GW (LIGO O4 freq.-dependent)	R27; R28	湧視/湧現	**Pass**	**Pass**
6	Quantum-enhanced bio microscopy (photodamage-limited)	Casacio et al., *Nature* 594, 201 (2021)	湧視/湧現	**Pass**	Cusp
7	N00N-state Heisenberg-limited interferometry	Afek, Ambar & Silberberg, *Science* 328, 879 (2010); Dowling, *Contemp. Phys.* 49 (2008)	湧視	**Pass**	Weak

#	Topic	Key Reference	Pillar	Phys. (i–ii)	Eng. (iii–iv)
8	Quantum illumination / target detection	Lloyd, *Science* 321, 1463 (2008); Tan et al., *PRL* 101, 253601 (2008)	湧視	Weak	Weak
9	Quantum OCT (dispersion cancellation)	Nasr et al., *PRL* 91, 083601 (2003)	湧視	**Pass**	Cusp
10	Entangled-photon ghost imaging	Pittman et al., *PRA* 52, R3429 (1995); Bennink et al., *PRL* 89, 113601 (2002)	湧現	Cusp	Weak
11	Hyperentanglement quantum holography	Defienne et al., *Nat. Photonics* 15, 104 (2021)	湧現/湧知	**Pass**	Cusp
12	Gaussian boson sampling / photonic quantum advantage	Zhong et al., *Science* 370, 1460 (2020) [Jiuzhang]; Madsen et al., *Nature* 606, 75 (2022) [Borealis]	湧現	**Pass**	Cusp
13	Fusion-based photonic QC (PsiQuantum Omega)	R29; R30	湧創/湧現	**Pass**	**Pass**
14	Large-scale integrated photonic quantum circuits (PICs)	R31; R32	湧創	**Pass**	**Pass**
15	Deterministic single-photon sources (QD-in-cavity)	R33; R34	湧創	**Pass**	**Pass**
16	Nonlinear metasurface entangled-photon sources	Ma et al., *eLight* 5, 2 (2025) — see R2; Santiago-Cruz et al., *Science* 377, 991 (2022)	湧現/湧視	**Pass**	Cusp
17	Quantum repeaters & quantum memories	Briegel et al., *PRL* 81, 5932 (1998); Duan, Lukin, Cirac & Zoller [DLCZ], *Nature* 414, 413 (2001); Sangouard et al., *RMP* 83, 33 (2011)	湧視	**Pass**	Cusp
18	Entanglement-enhanced optical atomic clocks	Pedrozo-Peñafiel et al., *Nature* 588, 414 (2020); Miklos et al., *Phys. Rev. Lett.* 135, 193202 (2025)	湧視/湧知	**Pass**	Cusp

Decomposed tally.

- **Physics inevitability (conditions i–ii):** 16 Pass, 1 Cusp, 1 Weak. Almost every topic has a rigorous classical asymptote and a credible scaling pathway.
- **Engineering inevitability (conditions iii–iv):** 6 Pass, 9 Cusp, 3 Weak. Only one-third of topics have closed the algorithmic-velocity and discovery→design-shift loop.
- **Double Pass (Barrett-inevitable):** 6 topics — items 1, 2, 5, 13, 14, 15. These are the only entries that pass all four conditions simultaneously.

This decomposition reveals a structural distinction that a single Pass/Cusp/Weak label would have obscured: quantum photonics is *physics-rich but engineering-cusp*. Almost every topic satisfies conditions (i)–(ii), but only a minority have crossed the engineering-inevitability threshold. The six Double-Pass entries are precisely those that have reached commercial or operational deployment: deployed QKD (items 1, 2), LIGO squeezed light in O4 (item 5), and manufacturable photonic hardware — PsiQuantum Omega chips, large-scale PICs, and deterministic quantum-dot single-photon sources (items 13–15). Quantum photonics therefore has a *broad-portfolio inevitability profile* rather than the *deep-single-topic inevitability* Barrett argues for digital quantum computing. The two profiles have distinct implications for research strategy, risk allocation, and public investment: a broad portfolio is more resilient to individual-topic setbacks but less likely to produce a single field-defining breakthrough.

Relation to Barrett. The decomposition exposes a subtle point Barrett's argument does not make explicit: physics inevitability and engineering inevitability are independent axes. Barrett's chemistry case passes both simultaneously (DFT fails *and* quantum computers are now being manufactured), so his implicit single-label test is adequate for that one case. Applied to a broader technology portfolio, the decomposition matters: a topic can be physics-inevitable yet engineering-stalled for decades — N00N-state interferometry (item 7) is the cleanest example, with a rigorous Heisenberg scaling limit (condition i) but no pathway beyond small photon number (condition ii fails in practice, conditions iii–iv

fail entirely). The Four-Pillar framework naturally captures both axes: conditions (i)–(ii) correspond to 湧視 (untapped physics), condition (iii) to 湧創 (continuous innovation via AI/algorithm loops), and condition (iv) to the 湧現 → 湧知 coupling already demonstrated in our cross-pillar linkage map.

5 Pillar 4: 湧知 (Emergent Knowledge) — New Pillar

Pillar 4: 湧知 — Emergent Knowledge

"New optical design knowledge that emerges from AI-human collaboration, where the AI discovers principles no human designer would have conceived."

#	Knowledge Domain	How It Emerges	Example / Evidence	Implication	Ref.
1	AI-discovered design topologies	LLM generates valid configs not in training data	OptiAgent: novel prescriptions via RL; outperforms traditional + vanilla LLM	New starting points human designers wouldn't explore	R1
2	Inverse design insights	NNs solve inverse scattering; reveal structure-function links	OptoGPT: dual-band AR coating, fewer layers, fraction of a second	Simpler solutions exposing hidden design rules	R10
3	Tolerance-aware principles	ML predicts mfg. sensitivity; reveals dominant parameters	ML identifies lens spacings as dominant tolerance drivers	"Optimize then tolerance" → "tolerance-informed optimization"	R11
4	PINN discoveries	PINN enforces physics; residuals reveal unmodeled physics	Singlet optimization + Eikonal Bridge; residuals flag model breakdown	PINNs as "physics debuggers" for model completeness	R21; R22
5	Cross-domain transfer	AI trained on one domain transfers to another	RL on aspheric singlet transfers strategies to other lens types	"Optical design intuition" in AI weights, extractable as heuristics	R15
6	Quantum-classical boundary	Metasurface quantum imaging reveals classical model limits	Multimodal quantum imaging with self-calibration	Quantitative thresholds for when quantum treatment is needed	R3; R4
7	Structured light generation	Spin-orbit coupling in simple materials	8 μm hBN: OAM ± 2 vortex via spin-orbit, no nanostructure	"Material-intrinsic" wavefront shaping without fab complexity	R7

6 Cross-Pillar Linkage Map

The four pillars are not independent—they form a coupled feedback system.

Connection	From	To	Mechanism	Status
Engineered emergence requires accessing hidden physics	湧視	湧現	Metasurfaces access spin-orbit, topology, quantum DOF → new emergent behaviors	Active
Continuous innovation generates new knowledge	湧創	湧知	Fast iteration loops produce unexpected solutions → new design principles	Emerging
Emergent knowledge reveals new physical frontiers	湧知	湧視	AI-discovered designs expose physics we didn't know was relevant	Early stage
New physics enables faster innovation cycles	湧視	湧創	Topology provides robust, fab-tolerant designs → fewer iterations needed	Theoretical
Emergence creates knowledge that feeds back	湧現	湧知	System-level surprises become codified design principles	Demonstrated (quantum self-cal.)
Knowledge accelerates engineered emergence	湧知	湧現	Understanding why emergence works allows designing for it intentionally	OptiAgent is first example

7 Key Reference Index

All references cited across the four pillars, with pillar relevance indicated. Click any reference number to jump to its full entry in Section 8.

#	Reference	Year	Pillar Relevance
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R2	Ma et al., "Quantum imaging using spatially entangled photon pairs from a nonlinear metasurface," eLight 5, 2	2025	湧現 + 湧視
R3	Zhu, Wang et al., "Metasurface-assisted multimodal quantum imaging," PNAS	2025	湧現 + 湧知
R4	"Quantum Imaging with Metasurfaces: Gains, Limitations, and Prospects," MDPI Photonics 13(1), 69	2026	湧視
R5	Penn State / Nanophotonics, "AI approach takes optical system design from months to milliseconds"	2026	湧創
R6	Lei et al., "Topological Quasiparticles of Light: Spin-Orbit Coupling to Photonic Skyrmions," Laser & Photonics Reviews	2025	湧視
R7	"Spin-orbit coupling in van der Waals materials for optical vortex generation," Light: Sci. & Appl.	2025	湧視 + 湧知
R8	Liu, Qiu & Yan, "Winding coupling phase for pseudo-spin-derived topological photonics," Nature Communications	2025	湧視
R9	"Model-free optical processors using in situ RL with PPO," Light: Sci. & Appl.	2026	湧創 + 湧知
R10	Ma, Wang & Guo, "OptoGPT: foundation model for inverse design in optical multilayer thin films," Opto-Electronic Advances 7, 240062	2024	湧知
R11	Yang et al., "Deep learning for polarization optical system automated design," Photonics 11	2024	湧創 + 湧知
R12	CES 2026 Optical and Lens Technology Depth Report, lensmanufacture.com	2026	湧創
R13	"AI Opens Up Optical System Design," Optica-OPN Vol. 37, January 2026	2026	All pillars
R14	LensAI, GitHub: HarrisonKramer/LensAI	Ongoing	湧創 + 湧知
R15	LensRL, GitHub: HarrisonKramer/LensRL	Ongoing	湧創
R16	ST Blog, "Top Tech Trends 2026: Metasurface technology shifts the paradigm"	2026	湧視
R17	Roadmap for Optical Metasurfaces, ACS Nano	2024	湧視 + 湧現
R18	Burkhardt et al., "Active Alignments of Lens Systems with Reinforcement Learning," arXiv:2503.02075	2025	湧創
R19	Hoshi et al., "Wavefront-aberration-tolerant diffractive deep neural networks using volume holographic optical elements," Scientific Reports 15, 1104	2025	湧現
R20	Yenra, "AI Optical System Design: 19 Advances (2025)," Review article	2025	湧現 + 湧創
R21	Raissi, Perdikaris & Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear PDEs," *J. Comput. Phys.* 378, 686	2019	湧知
R22	Chern, `github.com/jlchern-251016`: Eikonal Bridge / QFI / Thermal Photonics trilogy	Ongoing	湧知 + 湧視
R23	P. Barrett, "Useful quantum computing is inevitable—and increasingly imminent," *MIT Technology Review*	2025	湧視 (framework)
R24	Bennett & Brassard, "Quantum cryptography: Public key distribution and coin tossing," Proc. IEEE ICCSSP 1984; reprint *Theor. Comput. Sci.* 560	1984	Double-Pass (item 1)
R25	Liao et al., "Satellite-to-ground quantum key distribution," *Nature* 549, 43	2017	Double-Pass (item 2)
R26	Yin et al., "Satellite-based entanglement distribution over 1200 km," *Science* 356, 1140	2017	Double-Pass (item 2)

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#	Reference	Year	Pillar Relevance
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R28	Tse et al., "Quantum-Enhanced Advanced LIGO Detectors in the Era of Gravitational-Wave Astronomy," *PRL* 123, 231107	2019	Double-Pass (item 5)
R29	PsiQuantum Team, "A manufacturable platform for photonic quantum computing," *Nature* 641, 876	2025	Double-Pass (item 13)
R30	Bartolucci et al., "Fusion-based quantum computation," *Nat. Commun.* 14, 912	2023	Double-Pass (item 13)
R31	Wang et al., "Multidimensional quantum entanglement with large-scale integrated optics," *Science* 360, 285	2018	Double-Pass (item 14)
R32	Silverstone et al., "On-chip quantum interference between silicon photon-pair sources," *Nat. Photonics* 8, 104	2014	Double-Pass (item 14)
R33	Senellart, Solomon & White, "High-performance semiconductor quantum-dot single-photon sources," *Nat. Nanotechnol.* 12, 1026	2017	Double-Pass (item 15)
R34	Tomm et al., "A bright and fast source of coherent single photons," *Nat. Nanotechnol.* 16, 399	2021	Double-Pass (item 15)

8 Complete Reference List

Full bibliographic details for all references cited in the four-pillar tables and the Classical-Asymptote Test section. References are ordered by their in-text index (R1–R23). Click any reference number to return to the Key Reference Index.

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R7 (Authors), "Spin-orbit coupling in van der Waals materials for optical vortex generation," *Light: Science & Applications*, vol. 14, August 2025. https://doi.org/10.1038/s41377-025-01926-7

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Note: For references marked "(Authors)" where only partial author information was available from search results, full author lists should be verified from the original publications. The 6 Double-Pass entries in Table 1 (items 1, 2, 5, 13, 14, 15) are anchored via R24–R34. Cusp and Weak entries cite primary sources inline (author-year) to keep the formal bibliography focused on the Barrett-inevitable topics; readers are encouraged to verify each inline primary paper independently.

Context: Review of NCKU Talk (May 29, 2025) | Date: April 9, 2026 | Version: v14

Original three pillars: **湧現** (Emergence), **湧創** (Continuous Innovation), **湧視** (Unaddressed Limits)

Proposed fourth pillar: **湧知** (Emergent Knowledge) | New in v14: Classical-Asymptote Test